RAJALAKSHMI ENGINEERING COLLEGE

Dhanusheya G 251801053-2025-AIDS C2

Started on	Thursday, 27 August 2026, 8:28 AM
State	Finished
Completed on	Thursday, 17 September 2026, 9:41 AM
Time taken	21 days 1 hour
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int majorityElement(int nums[], int left, int right) {
    // Base case: only one element
    if (left == right) {
        return nums[left];
    }

    // Divide
    int mid = (left + right) / 2;

    // Conquer
    int leftMajority = majorityElement(nums, left, mid);
    int rightMajority = majorityElement(nums, mid + 1, right);

    // If both halves have the same majority
    if (leftMajority == rightMajority) {
        return leftMajority;
    }

    // Count both candidates in current range
    int leftCount = 0;
    int rightCount = 0;

    for (int i = left; i <= right; i++) {
        if (nums[i] == leftMajority)
            leftCount++;

        if (nums[i] == rightMajority)
            rightCount++;
    }

    // Combine
    if (leftCount > rightCount)
        return leftMajority;
    else
        return rightMajority;
```

```
    return rightMajority;
}

int main() {
    int n;
    scanf("%d", &n);

    int nums[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }

    int result = majorityElement(nums, 0, n - 1);

    printf("%d\n", result);
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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